

SAHIL DUWAL

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SUMMARY

Highly motivated Computer Engineer with significant experience in AI, Machine Learning, and Web Development. Proficient in designing intelligent deep learning architectures, computer vision algorithms, and resilient full-stack web applications. Passionate about applying cutting-edge AI technology to build efficient, impactful digital products.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, C, C++, HTML5, CSS3

AI, Machine Learning & Vision: PyTorch, Computer Vision, Deep Learning, Natural Language Processing (NLP), scikit-learn

Full-Stack Web Development: Node.js, React, Next.js, FastAPI, Tailwind CSS, RESTful APIs

Data Science & Databases: Pandas, NumPy, Matplotlib, MySQL, Data Analysis, Statistical Modeling

Developer Tools & Frameworks: Git, Linux/Unix, Git Workflow, LaTeX

SOFT SKILLS

Technical Communication • Collaborative Problem Solving • Analytical Thinking • Adaptability • Project Management

EXPERIENCE

Full Stack Development Trainee

May 2026 – July 2026

CVArcher

- Engineered responsive web applications utilizing TypeScript and Node.js for robust server-side logic and type-safe development.
- Integrated secure user authentication systems leveraging Google OAuth and implemented reliable payment gateway integration.
- Conducted rigorous quality assurance (QA) and functional testing to optimize application performance, eliminate bugs, and improve system reliability.

Data Annotation Intern

October 2025 – December 2025

Next Wave AI

- Annotated high-quality image datasets (including objects and media covers) to support the training and development of computer vision models.
- Generated structured JSON schemas for AMS360 to facilitate seamless workflow automation.
- Performed comprehensive website reviews and functional testing to identify UI/UX bottlenecks and performance issues.
- Evaluated Large Language Model (LLM) security behaviors and vulnerabilities through targeted prompt testing and adversarial analysis.
- Researched and conducted hands-on capability assessments of emerging AI video generation tools.

DataCamp Fellowship

2025 – 2026

Code for Nepal

- Selected for an intensive data science fellowship program focused on advanced analytical methodologies.
- Accelerated technical expertise in Python, exploratory data analysis (EDA), and practical machine learning applications.

Tech Support Member – Hult Prize On-Campus Program

2024 – 2025

Khwopa College of Engineering

- Delivered comprehensive technical support for event management, complex presentation setups, and digital logistics.
- Assisted participating teams in troubleshooting IT-related and software issues during high-stakes workshops, mentoring sessions, and hackathons.
- Ensured flawless technical operations across multiple campus-wide events, directly contributing to the program's operational success.

EDUCATION

Bachelor of Computer Engineering

2022 – 2026

Khwopa College of Engineering, IOE – Libali, Bhaktapur

- Pursued a comprehensive program emphasizing Artificial Intelligence, Machine Learning, and Software Engineering.
- Engaged in practical projects and research focused on intelligent systems and real-world problem solving.

CERTIFICATES & COURSES

- Learn HTML – Programiz Pro | 2024
- SQL Basics – Programiz Pro | 2024
- Learn Python Basics – Programiz Pro | 2024
- Python Essentials I – Cisco Networking Academy | 2024
- Introduction to Cybersecurity – Cisco Networking Academy | 2024
- Introduction to IoT – Cisco Networking Academy | 2024
- AI Fundamentals – DataCamp | 2025
- Data Skills for Business – DataCamp | 2025
- Understanding Data Topics – DataCamp | 2025
- EU AI Act Fundamentals – DataCamp | 2025

ACHIEVEMENTS

Hult Prize Nepal	2022 – 2023
• Reached semi-finals with a campus-team startup idea, presenting an innovative social impact solution.	
CodeYatra Hackathon	2025
• Participated in a national-level innovation hackathon focused on AI-driven problem-solving.	
Locus Pattern Verse	2025
• Advanced to Phase 2 in a national tech competition with an AI-powered rug-design generation system.	

PROJECTS

AI, Machine Learning & Data Science

Deepfake Detection & Localization

Developed a hybrid ViGNN-U-Net model for detecting and localizing deepfake content with high accuracy and robustness across various manipulation types.

Tech: PyTorch, Computer Vision, Deep Learning, U-Net, ViGNN

Quantum-Enhanced BERT Model — [View Repository](#)

Hybrid classical-quantum ML pipeline combining multilingual BERT embeddings with variational quantum circuits (VQCs) for low-resource Nepali offensive-speech detection.

Tech: PyTorch, BERT, Variational Quantum Circuits, NLP

Shruti: AVSR in Nepali Language — [View Repository](#)

Audio-visual speech recognition (AVSR) system built for the Nepali language, combining acoustic and visual cues for improved recognition accuracy.

Tech: Deep Learning, Computer Vision, NLP, Python

Ghumti: AI Travel Assistant

Built an intelligent chatbot powered by Llama 3 to provide real-time local bus routes, schedules, and optimized travel information.

Tech: Llama 3, NLP, Chatbot, API Integration

PDF Q&A RAG Bot — [View Repository](#)

Local retrieval-augmented generation (RAG) application enabling semantic Q&A across multiple PDFs, running entirely offline.

Tech: Llama2, LangChain, DocArray, Sentence-Transformers

Captioning Image — [View Repository](#)

Image captioning model combining a pretrained ResNet50 encoder with an LSTM decoder to generate descriptive captions for input images.

Tech: PyTorch, ResNet50, LSTM

Web & Software Development

CentraSpend – Expense Tracker — [View Repository](#)

Full-stack personal expense tracking application with account management and transaction history.

Tech: React, Express, Node.js, MongoDB

RecipeBook: Cooking Recipe Portal — [View Repository](#)

Web application for storing, organizing, and managing cooking recipes with a full client-server architecture.

Tech: *React, FastAPI, MySQL*

Personal Portfolio Website — [Portfolio 2.0](#)

Personal portfolio website showcasing skills, experience, and completed projects; rebuilt as a v2 with an updated design.

Tech: *TypeScript, React, CSS*